

CS 621 PROJECT — YOUTUBE COMMENT ANALYSIS

Voices in the Comments

What 20,080 YouTube comments reveal about two gaming audiences

● dunkey · 7.57M subscribers

● Game Grumps · 5.45M subscribers

Two channels, one question

Comment sections are the most direct record of how an audience feels. **We compare the top comments on each channel's most-viewed videos.**

Do a review channel and a personality channel earn different kinds of audience conversation — in sentiment, timing, and vocabulary?

Hypothesis: dunkey's audience reacts to videos (front-loaded, quotable), Game Grumps' audience talks about the hosts (steady, personal).

 **dunkey**

7.57M subscribers

Comedy game reviews · 995 uploads

10 most-viewed videos analyzed

 **Game Grumps**

5.45M subscribers

Let's-play duo · 9,881 uploads

10 most-viewed videos analyzed

Pipeline and data source

1. Ingestion

2. Engineering

3. Analytics

4. Visualization

5. Delivery

Source — YouTube Data API v3. For each channel we rank the full uploads playlist by actual view count (the API's own “order by views” search is unreliable on large channels), then page through each top video's comment threads, **ordered by relevance** — the highly-liked comments that define its conversation. Every raw response is stored as JSON before any processing.

20,080

raw comments collected

20

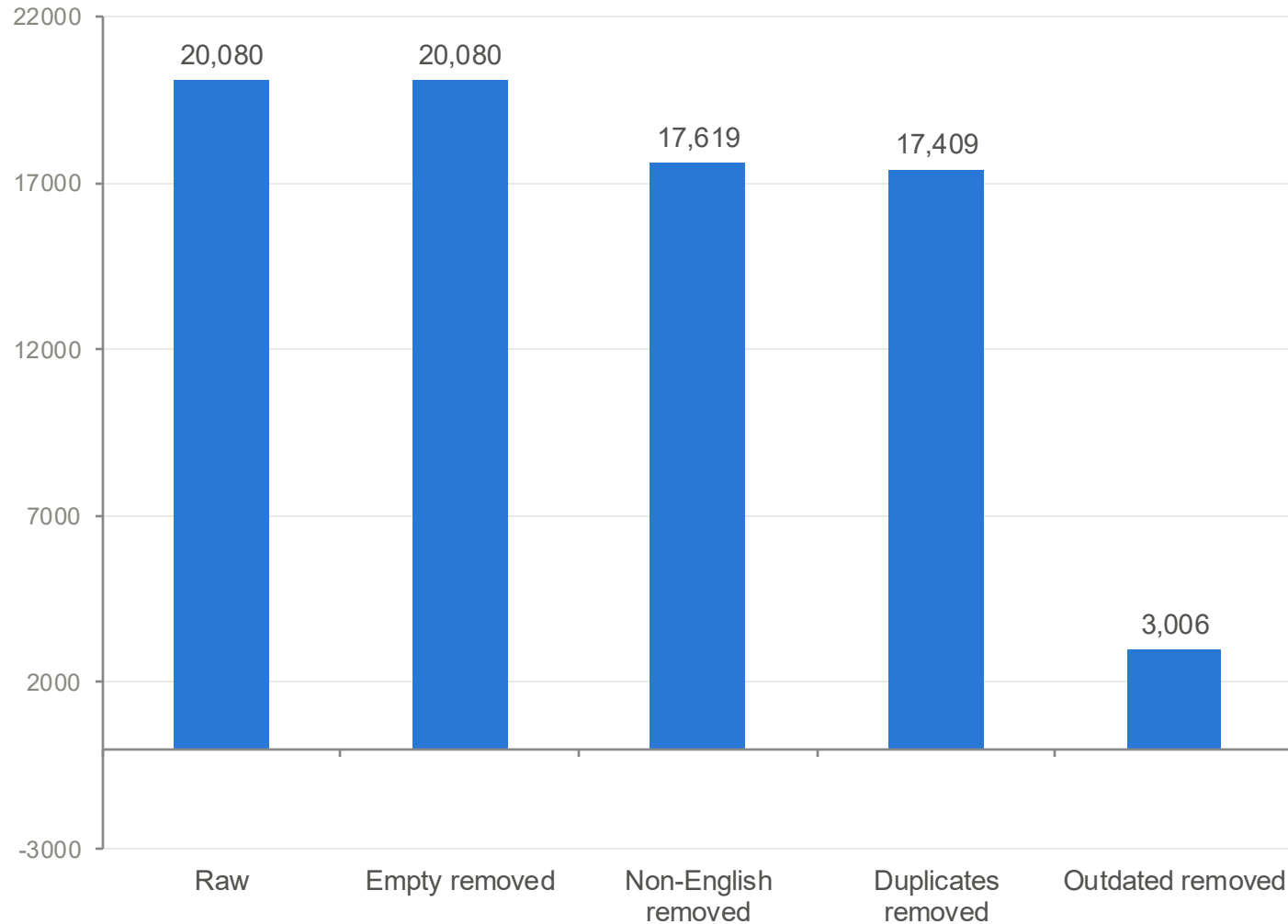
videos across 2 channels

20

JSON files (1 per video)

Each record: comment ID, video ID, author, text, like count, timestamp — held in lists/dicts at collection, then pandas Series and DataFrames.

Data engineering: four cleaning rules



1 • Empty

Whitespace-only text. Real finding: YouTube blocks empty posts — 0 of 20,080 removed.

2 • Non-English

langdetect (seeded for reproducibility) drops ~12% so one sentiment model fits the corpus.

3 • Duplicates

Same text on the same channel: copy-paste and bot spam.

4 • Outdated

Keeps each video's first 18 months — its reception window — rather than a calendar cutoff, so older classics aren't unfairly stripped.

20,080 → 3,006 comments · the reception-window rule does most of the work

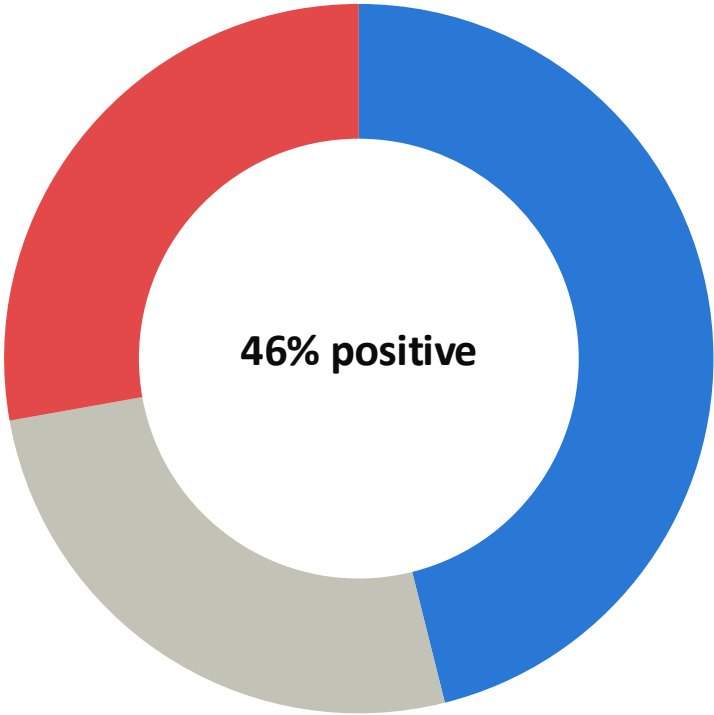
The cleaned dataset, side by side

	dunkey	Game Grumps
Cleaned comments	1,262	1,744
Unique commenters	1,223 (97%)	1,699 (97%)
Mean comment length	102 chars	90 chars
Median likes per comment	120	19
Comments containing emoji	2%	2%
Videos represented	10	10

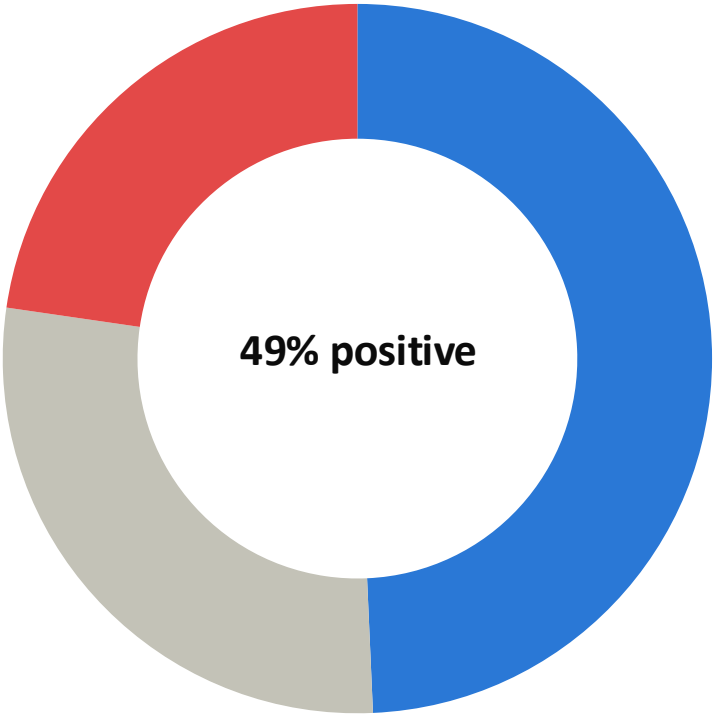
What jumps out

- Both comment sections are crowds, not clubs — 97% of top comments come from different people.
- dunkey's top comments earn 6× the median likes (120 vs 19): viral videos concentrate applause on a few one-liners.
- Emoji are rare at the top (~2%): the most-liked comments are written jokes, not reactions.

Sentiment:



● **dunkey**



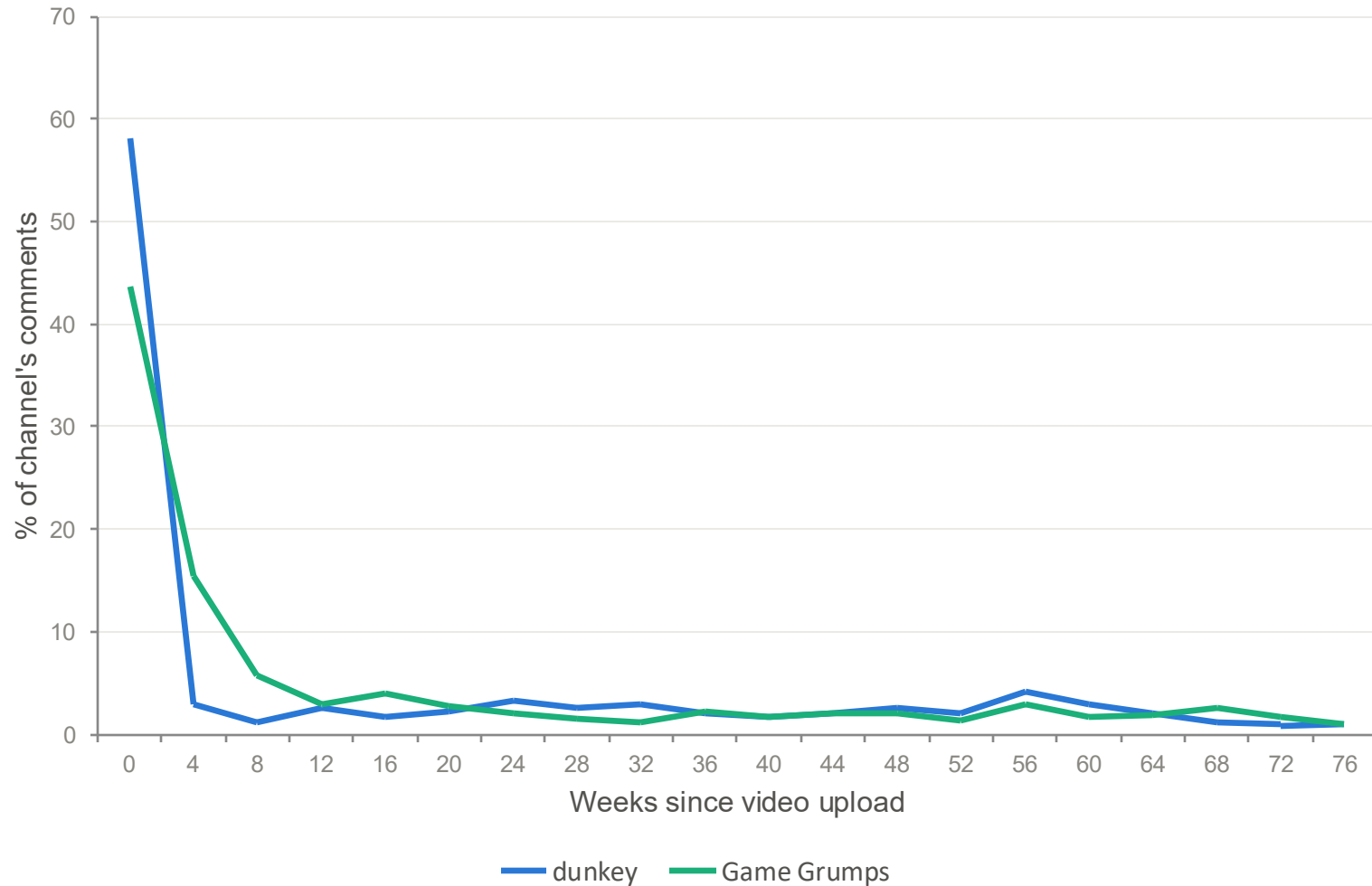
● **Game Grumps**

● Positive ● Neutral ● Negative

VADER polarity on every cleaned comment.

Game Grumps runs warmer (49.3% vs 46.1% positive; 22.7% vs 27.8% negative): comments about beloved hosts skew kind.

When the conversation happens



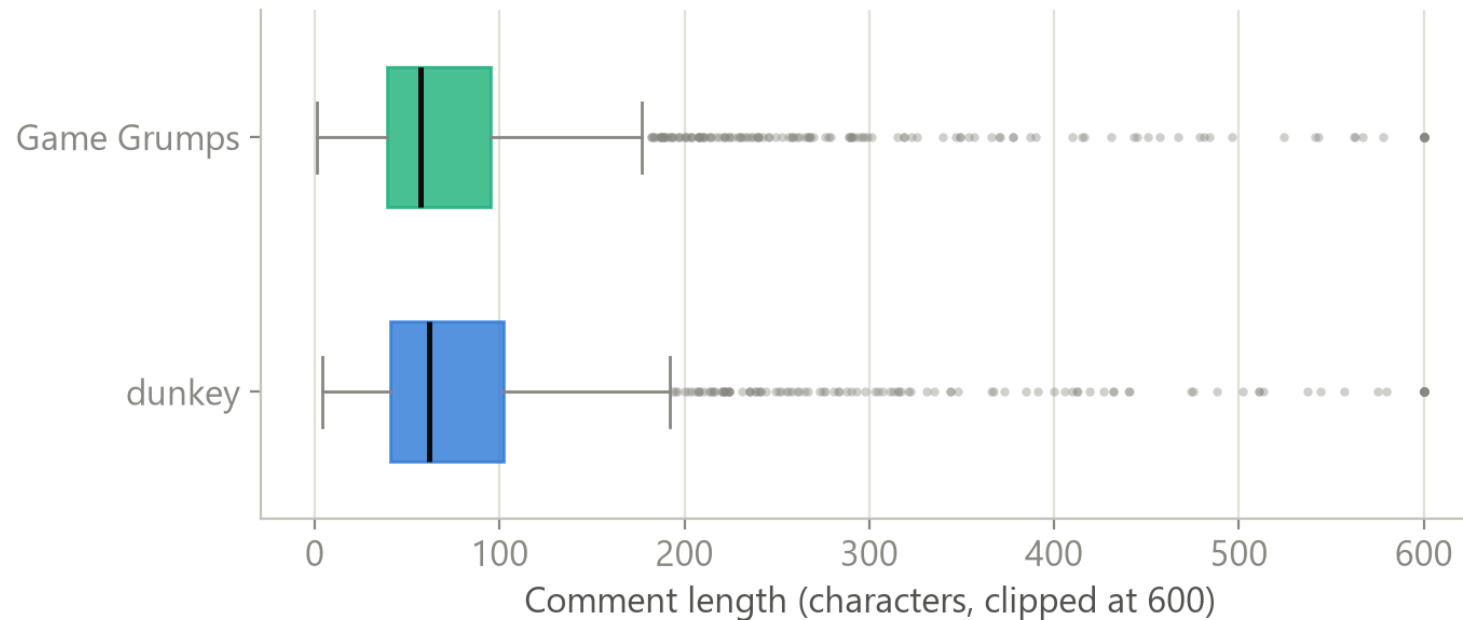
dunkey:

58% of top comments land in the first 4 weeks — the audience swarms each release, then moves on.

Game Grumps:

44% arrive in the first month, but a quarter of the conversation still happens 6+ months in, as fans revisit classic moments.

How much do they write?



Boxplot of comment length; dots are outliers past the whiskers (clipped at 600 characters).

102 vs 90

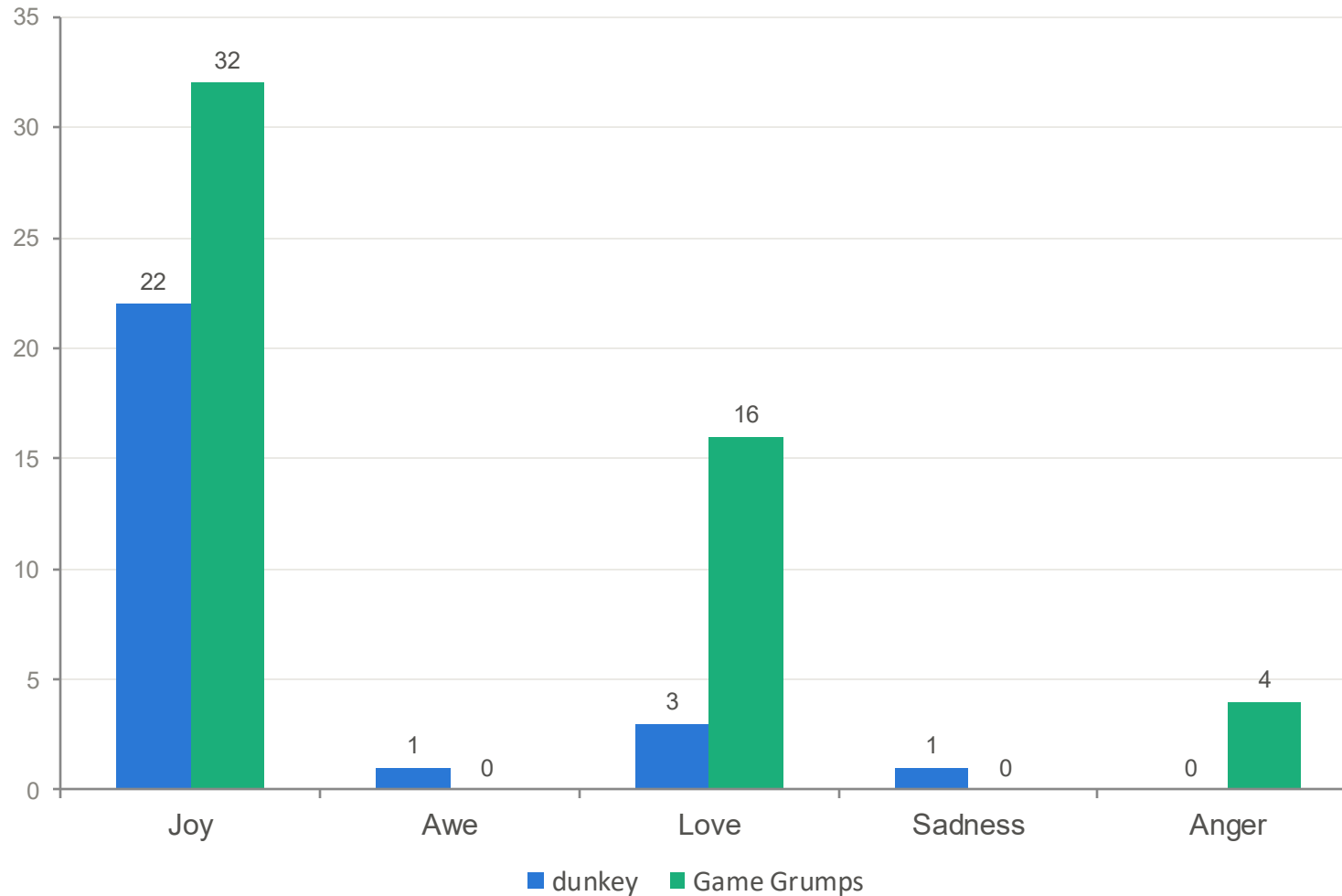
mean characters (dunkey vs GG)

Essays exist

both tails reach 600+ characters

Median comments are one-liners on both channels; dunkey draws slightly longer, more essay-like comments.

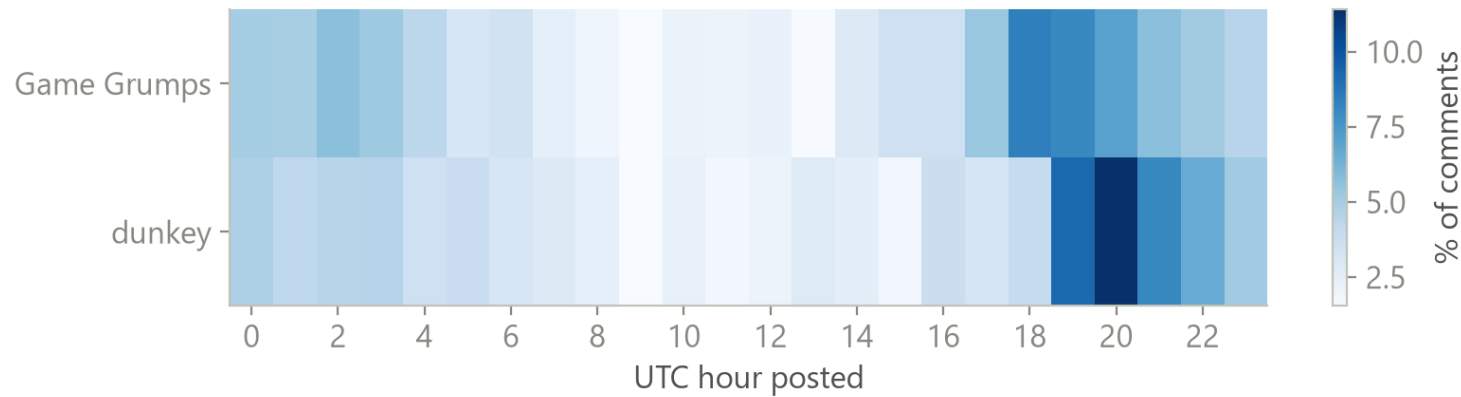
Emoji emotion analysis



- Joy dominates both channels — these are comedy audiences.
- Game Grumps shows 5× the love emoji (16 vs 3): affection for the hosts, not just the videos.
- Only ~2% of top comments use emoji at all — highly-liked comments are crafted text.

Emoji mapped to emotions (😂→joy, 🤯🔥→awe, ❤️→love, 😞→sadness, 😡→anger)

Where (roughly) is the audience?



Peak hour (UTC)

dunkey — 20:00

Game Grumps — 18:00

Share of each channel's comments by UTC hour of posting.

The API exposes no commenter location, so we triangulate:

- Posting hours peak at 18–20 UTC for both channels.
- The ~12% of comments removed as non-English — their language mix is itself the geographic signal.
- *Honest label: this is a proxy, not measurement. True spatial analytics would need data the API doesn't provide.*

Two vocabularies, two relationships



dunkey's cloud is about the games: dark souls, graphics, music, mario — review vocabulary. **Game Grumps' cloud is about the people:** arin, grumps, show, missed — the hosts are the content.

What the comments told us

Same warmth, different target

Both ~50% positive, but Game Grumps runs warmer and its love flows to the hosts; dunkey's applause goes to the videos.

Premiere vs. rewatch

58% of dunkey's conversation lands within a month of upload; Game Grumps sees a quarter of its conversation arrive 6+ months in.

Crowds, not clubs

97% unique commenters — top comments are drive-by wit, and emoji-free craftsmanship wins the likes.

Deliverables

Raw JSON dataset (20,080 comments · 20 videos) · cleaned dataset (3,006 rows) · written report (PDF) · this deck · documented Python pipeline (~330 lines)